

Employee Directory Report

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Working UI Screenshot

The application displays the add/edit form, search and sort controls, seeded employee records, active headcount, and active/inactive status badges.

Admin Console
HR Department

HR Portal | Dashboard | **Directory** | Analytics | Reports

Employee Directory
Manage staff records with Vue, Axios, Express and MySQL.

Summary:
 TOTAL EMPLOYEES: 21
 ACTIVE EMPLOYEES: 19
 INACTIVE EMPLOYEES: 2

NEW RECORD: Add Employee

Employee ID: EMP001

Full Name: Aina Sofia Rahman

Email: name@company.my

Department: Select department

Position: Systems Analyst

Hire Date: dd-yy-yy | Salary (RM): 1500

Search and Sort:
 Search: Name, ID, email or department
 Sort by: Name | Order: Ascending | Reset

Table:
 Page 1 of 3 | 21 records | Previous | Next

EMPLOYEE	DEPT & POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
AS Aina Sofia Rahman EMP001 - aina.rahman@company.my	HR HR Executive	2021-03-15	RM 4,200.00	Active	Edit Delete
AH Amirul Hakim EMP011 - amirulhakim@company.my	IT Database Administrator	2019-04-08	RM 7,600.00	Active	Edit Delete
CP Chong Pei Shan EMP010 - pei.chong@company.my	Marketing Content Strategist	2020-02-18	RM 5,600.00	Active	Edit Delete

Figure 1: Vue Employee Directory running at <http://127.0.0.1:5174> with MySQL seed data.

Chapter 8 Learning Outcomes

- **LO1: Connect Vue to a backend.** The Vue app calls the Express REST API on port 3001 and keeps the employee list in the parent component state.
- **LO2: Use Axios.** All HTTP requests go through one Axios instance in `src/services/api.js`, including a request logger and a response error mapper.
- **LO3: Build validated Vue forms.** `EmployeeForm.vue` uses `v-model.trim` and `v-model.number`, validates all required fields, and displays inline errors before any request is sent.
- **LO4: Perform CRUD with MySQL.** Express implements create, read, update, delete, server-side search, and whitelisted sorting with prepared statements through the `mysql2` promise pool.

Implementation Notes

The application is separated into form, list, and search/sort components. The SQL script creates the database, table, and 21 sample records across HR, IT, Finance, Marketing, and Operations, with inactive employees included for status testing.

Challenges and Resolutions

- **Safe sorting:** Directly accepting a column name from the client would risk SQL injection. The API maps requested sort keys to a fixed whitelist before building the ORDER BY clause.
- **Consistent errors:** Network and server errors can have different Axios shapes. The response interceptor converts them into one message format used by the UI banner.
- **Responsive layout:** The form and directory use CSS grid and a horizontally safe table layout so the page remains usable on laptop and tablet widths.